Ion-model calibration and coordination

An NMDAR-motivated Ca2+/Mg2+ benchmark in TIP3P water and acetamide

Xiangpu Wang | Completed local computational study | 72.0 ns of production across 36 cases

Question. How much do two published ion calibrations change local hydration and finite-time amide contact, before attempting an NMDAR pore calculation?

Observed. IOD-minus-CM mean RDF-peak shifts were +0.0040 nm for Mg and -0.0040 nm for Ca. The table records between-replica variation and initial-state contrasts; it makes no significance or affinity claim.

Group	RDF peak, nm (mean +/- SD)	Water CN (mean +/- SD)	Amide contact-start minus separated-start fraction (percentage points)
Mg / CM	0.2030 +/- 0.0000	6.00 +/- 0.00	+100.0
Mg / IOD	0.2070 +/- 0.0000	6.00 +/- 0.00	+100.0
Ca / CM	0.2483 +/- 0.0012	8.04 +/- 0.01	+66.3
Ca / IOD	0.2443 +/- 0.0012	8.00 +/- 0.01	+20.6

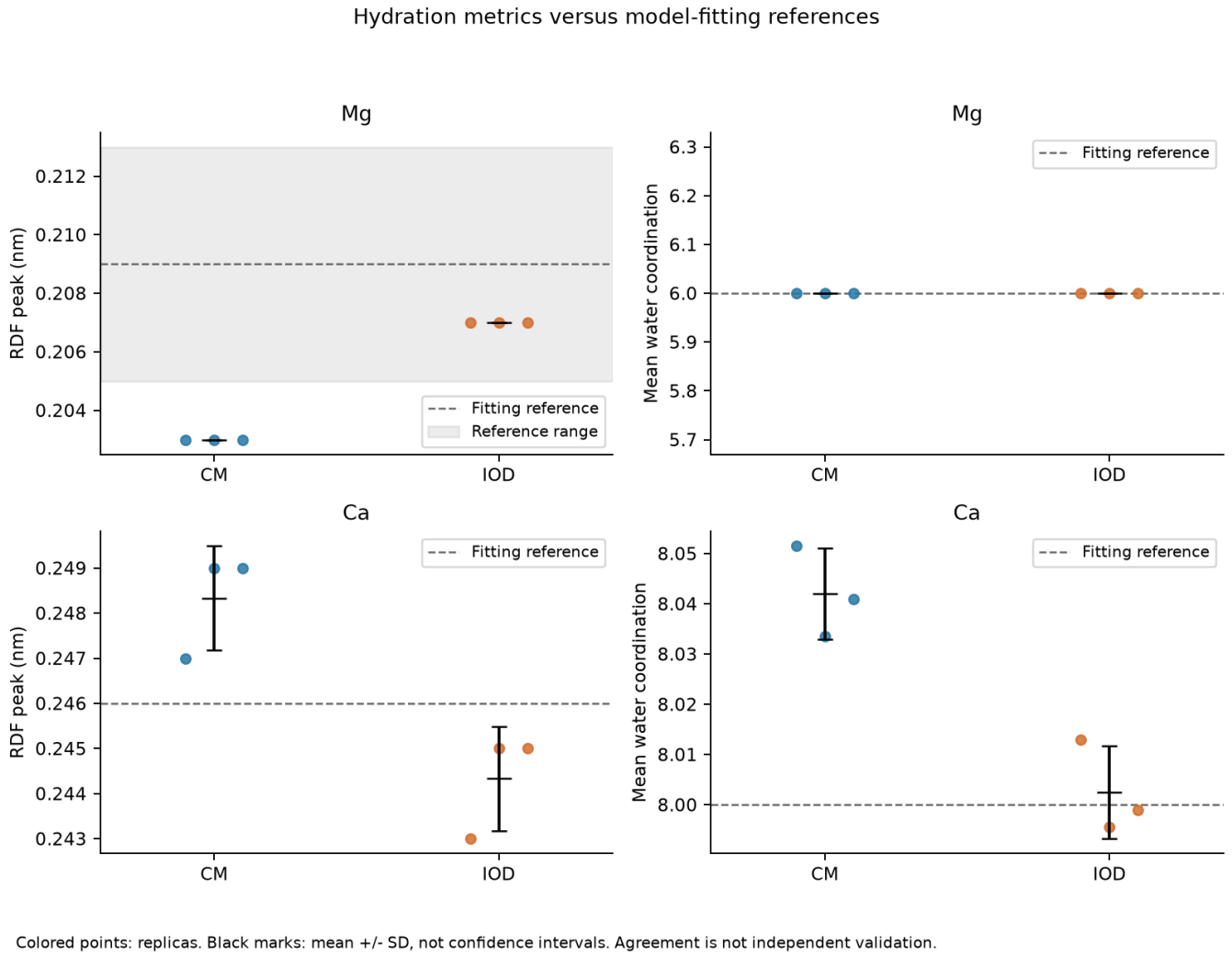


Figure 1. Hydration metrics and fitting references [1,2]. Three points per group are separate simulations; SD is not a confidence interval. Amide contact uses the diagnostic water-shell cutoff and is defined on page 5.

Methods and parameter provenance

The same full-charge (+2 e) Ca2+ and Mg2+ models were compared in standard TIP3P water. CM is Li-Merz's compromise 12-6 set; IOD is the distance-targeted set [1]. Lorentz-Berthelot mixing and PME were retained. No charge scaling or 12-6-4 term was introduced.

Item	Recorded setup
Design	2 ions x 2 models x 3 replicas; plus contact/separated amide starts (24 cases)
Temperature / pressure	298.15 K / 1.00 bar
Initial periodic cube / counterions	3.00 nm / one cation + two chloride ions; no extra salt
Finite-concentration system	Cation 0.0622 to 0.0631 M across case means from actual volumes; chloride is twice this concentration
Integrator	Langevin-middle; step 2.0 fs; friction 1.00 per ps
Constraints / pressure control	Rigid water and H-bond constraints; isotropic Monte Carlo barostat every 25 steps
PME / nonbonded cutoff	Error tolerance 0.00010; real-space cutoff 1.00 nm
Equilibration / saved cadence	100.00 ps / 1.00 ps
Production per case	Hydration 2.00 ns; amide 2.00 ns

Group	Amber Rmin/2 (angstrom)	epsilon (kcal/mol)
Mg / CM	1.360	0.01020237
Mg / IOD	1.395	0.01491700
Ca / CM	1.649	0.10592870
Ca / IOD	1.608	0.08337961

Replicas share a solvent-packing template but use separately seeded velocities, barostat moves and chloride placement. They test stochastic repeatability of this preparation; they do not independently sample every possible initial hydration state. The approximately 0.063 M mean cation concentration is finite, not infinite dilution.

Acetamide. FreeSolv mobley_8048190 supplies GAFF/AM1-BCC parameters [3,4]. Its neutral, nine-atom topology was converted using OpenMM's Amber reader, preserving exclusions and 1-4 scaling. Carbonyl oxygen O1 was initially placed in direct contact or at 0.65 nm; each start was then minimized and equilibrated. These are preparation labels, not enforced production states.

Conversion check. The source geometry and four perturbed geometries passed independent source-topology versus XML comparisons. Maximum per-force energy and atomic-force differences were 1.02e-07 kJ/mol and 5.02e-06 kJ/mol/nm. This verifies format conversion, not metal-amide accuracy.

Exact source commits, original files and SHA-256 checksums are retained in SOURCES.md and parameters/source_manifest.json. All completed cases carry the protocol, code hash, initial/final structures and runtime metadata.

Hydration structure: the measured comparison

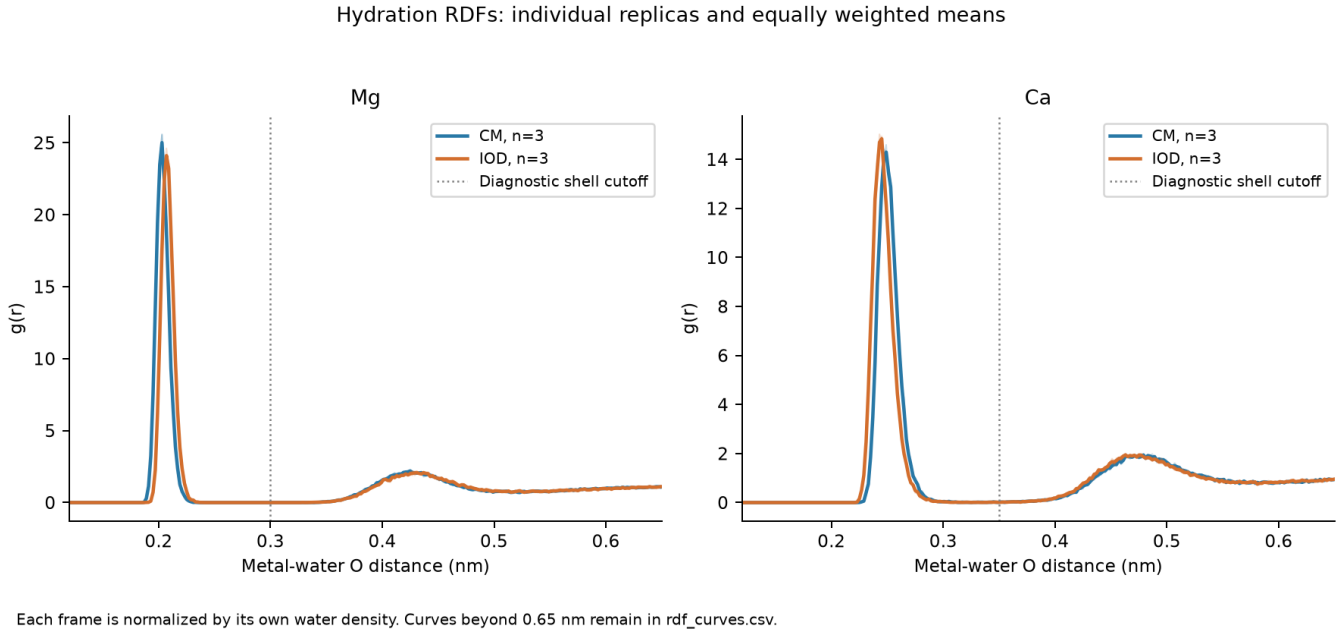


Figure 2. Individual-replica metal-water oxygen RDFs and equally weighted group means. Each frame's histogram is normalized by its own water number density. The CSV retains the full 0-0.9 nm range; histogram bins are 0.002 nm.

Group	Pooled in-shell mean distance (nm)	Mean water CN	Counterion contact fraction (%)
Mg / CM	0.204269 +/- 0.000031	6.00 +/- 0.00	0.4 +/- 0.3
Mg / IOD	0.208612 +/- 0.000032	6.00 +/- 0.00	0.1 +/- 0.1
Ca / CM	0.252530 +/- 0.000119	8.04 +/- 0.01	0.0 +/- 0.0
Ca / IOD	0.247932 +/- 0.000065	8.00 +/- 0.01	0.0 +/- 0.0

The pooled in-shell mean distance and the RDF peak are different estimators. The mean conditions on all water oxygens inside the declared shell; the peak is the maximum RDF bin within the predeclared ion-specific window. A peak value is therefore discrete at this bin width, and a small replica SD does not eliminate binning or model uncertainty.

The reference peak distances are 0.209 +/- 0.004 nm for Mg and 0.246 nm for Ca; reference coordination values are 6 and 8 [2]. These values belong to the calibration lineage. Any agreement with these calibration references constitutes target recovery, not independent predictive validation. No uncertainty is invented for the Ca target or the coordination values.

Water-shell cutoffs are 0.30 nm for Mg and 0.35 nm for Ca. Counterion contact is the fraction of saved frames with the nearest chloride below 0.40 nm. A counterion contact can alter the local environment and is reported rather than silently discarded. Acetamide was absent from these hydration comparisons.

Sampling diagnostics and definition sensitivity

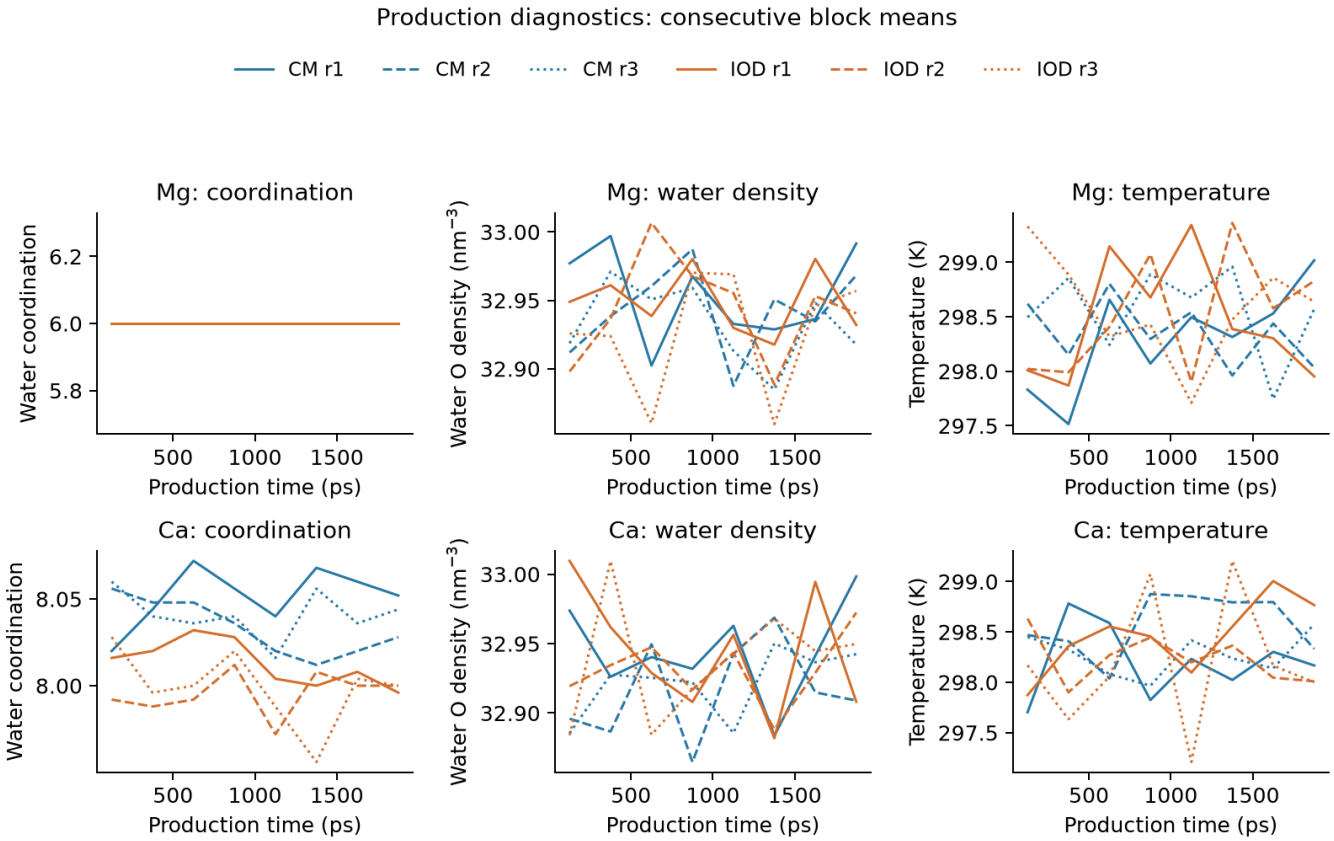


Figure 3. Consecutive block means in the designated production segment. Blocks display time dependence; they are not additional independent replicas. No automatic trimming was used to improve agreement.

Group	Second-half minus first-half CN (replica range)	Temperature half-change (K; range)	CN span across cutoff +/-0.02 nm (range)
Mg / CM	0.000 to 0.000	-0.22 to 0.57	0.000 to 0.000
Mg / IOD	0.000 to 0.000	-0.32 to 0.29	0.000 to 0.000
Ca / CM	-0.027 to 0.007	-0.04 to 0.25	0.037 to 0.043
Ca / IOD	-0.024 to -0.001	-0.16 to 0.29	0.030 to 0.038

Across hydration replicas, the largest absolute half-to-half CN change was 0.027; the largest cutoff-dependent CN span was 0.043. These are diagnostic magnitudes, not a convergence test with a universal acceptance threshold.

In 6 of 12 hydration trajectories, no first-shell membership change was observed between saved frames. A fixed coordination number can conceal water replacement, and unchanged water identities may reflect an unsampled slow process. Saved-frame observations do not establish committed exchange events or residence times.

Three independently seeded preparations provide descriptive repeatability checks. All replica values and the sample SD are reported, without independent-frame confidence intervals or p-value-based rankings. Shared preparation procedures and unvisited states can still limit independence and coverage [5].

Acetamide: finite-time initial-state contrasts

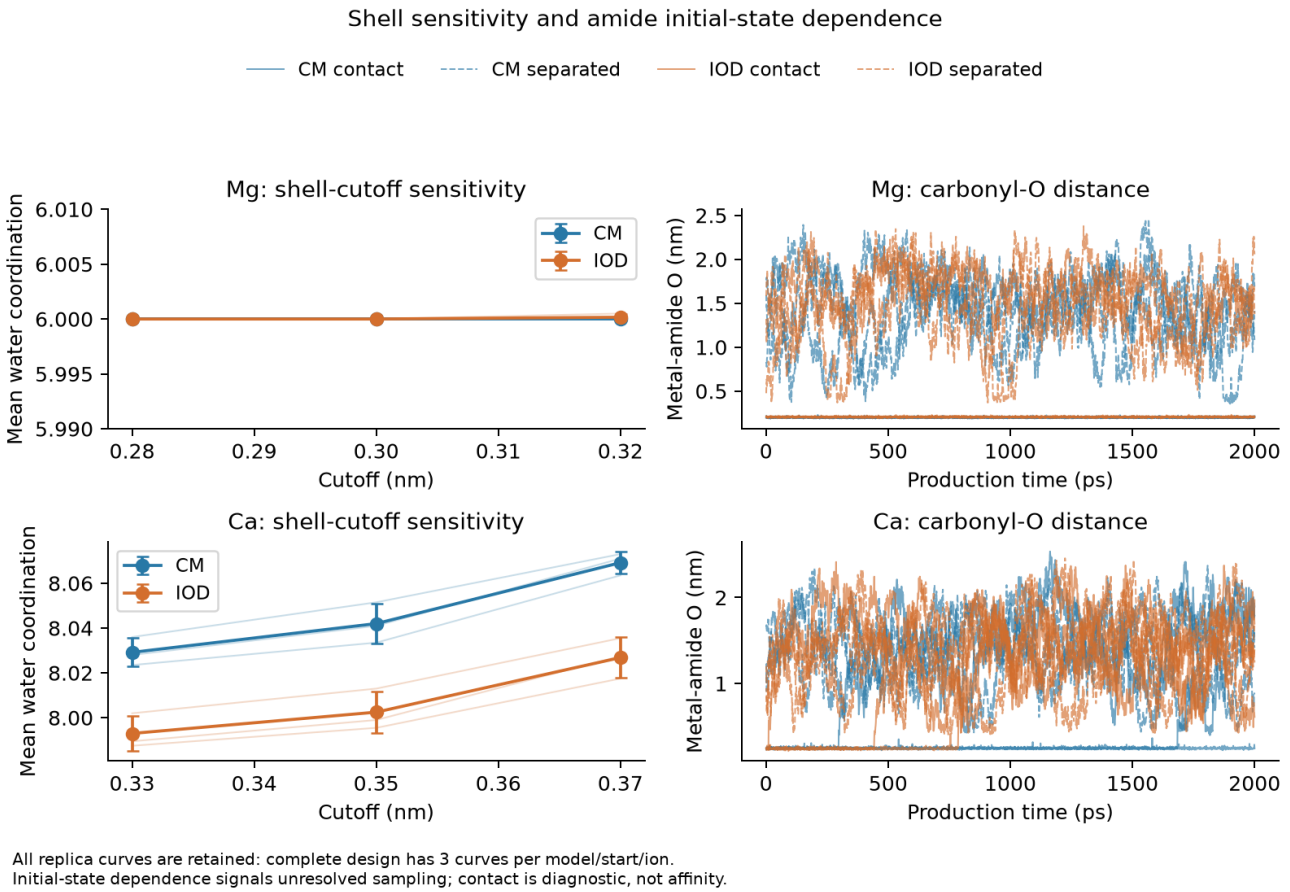


Figure 4. Left: hydration shell-cutoff sensitivity. Right: metal-carbonyl oxygen distance in the 24 amide simulations. Solid/dashed preparation labels denote contact/separated starts; replicas are separate trajectories.

Group	Prepared start	Mean carbonyl-O distance (nm)	Diagnostic contact fraction (%)
Mg / CM	contact	0.20083 +/- 0.00009	100.0 +/- 0.0
Mg / CM	separated	1.41965 +/- 0.04543	0.0 +/- 0.0
Mg / IOD	contact	0.20500 +/- 0.00005	100.0 +/- 0.0
Mg / IOD	separated	1.48900 +/- 0.10009	0.0 +/- 0.0
Ca / CM	contact	0.65704 +/- 0.59039	66.3 +/- 45.3
Ca / CM	separated	1.40881 +/- 0.09558	0.0 +/- 0.0
Ca / IOD	contact	1.15341 +/- 0.27092	20.6 +/- 19.4
Ca / IOD	separated	1.37190 +/- 0.19314	0.0 +/- 0.0

Diagnostic contact is carbonyl oxygen below 0.30 nm (Mg) or 0.35 nm (Ca), reusing the water-shell cutoff as an explicit operational definition. Each table entry is the mean +/- SD across three replicas. Neither the cutoff nor a contact-start trajectory defines a calibrated metal-amide bound state.

Contact-start minus separated-start fraction contrasts (percentage points): Mg/CM +100.0; Mg/IOD +100.0; Ca/CM +66.3; Ca/IOD +20.6. Initial-state contrasts reveal finite-time sensitivity; similar fractions alone would not establish equilibration. No equilibrium affinity, free-energy difference or kinetic rate is inferred.

Interpretation, reproducibility and references

Main finding. 12 of 12 separated-start trajectories showed no saved-frame amide contact. Mg contact-start minus separated-start contrasts were +100.0 percentage points with CM and +100.0 with IOD. Ca contact-start fractions averaged 66.3 +/- 45.3% with CM and 20.6 +/- 19.4% with IOD (mean +/- replica SD). The large Ca spread and persistent preparation dependence are the central sampling result; these contact fractions do not rank equilibrium affinities.

Contribution. This is a completed, reproducible small-system comparison of calibration choices and local coordination observables. It supplies measured hydration differences and an amide sampling diagnostic relevant to planning an Asn-associated ion model. It does not simulate an NMDAR receptor or establish its pore mechanism.

Next discriminating calculation. Establish reversible sampling and an external metal-amide benchmark before using these combinations for quantitative binding claims. A later pore study would additionally need construct-specific geometry, an adequate ion/protein interaction model and a converged multidimensional sampling strategy. The current data do not select a universal winning force field.

Measured feasibility. Production throughput across all 36 cases was 262.4 ns/day (total simulated time divided by summed production wall time); case-specific values spanned 112.8 to 508.4 ns/day. Summed case wall time was 7.02 hours. Pilot throughput is excluded. Timing includes concurrent desktop/GPU use and should not be treated as isolated hardware performance. Device: NVIDIA GeForce RTX 3050 Laptop GPU. OpenMM: 8.6.1.dev-b399af4.

Reproduction. Run simulate.py for hydration and amide with the committed protocol, then analyze.py with both campaigns, and build_report.py. The builder needs ReportLab; its separate project Python runs the mandatory verify_project.py data audit. Simulation dependencies are in requirements.txt. Raw NPZ/DCD/checkpoint files remain local and Git-ignored; tables, figures, manifests and scripts retain the audit record. Final generation requires 36 complete cases, raw-sample hashes and summaries recomputed from raw data.

Generative-AI contribution. Codex assisted project scoping, source checking, code authoring, analysis and report drafting. Numerical findings are computed from the saved molecular-dynamics outputs; generated text does not replace parameter provenance or the independent checks. This assistance should remain disclosed when the work is shown to a supervisor.

References

[1] Li P, Roberts BP, Chakravorty DK, Merz KM Jr. Rational Design of Particle Mesh Ewald Compatible Lennard-Jones Parameters for +2 Metal Cations in Explicit Solvent. J Chem Theory Comput. 2013;9:2733-2748. 10.1021/ct400146w

[2] Li Z, Song LF, Li P, Merz KM Jr. Systematic Parametrization of Divalent Metal Ions for the OPC3, OPC, TIP3P-FB, and TIP4P-FB Water Models. J Chem Theory Comput. 2020;16:4429-4442. 10.1021/acs.jctc.0c00194

[3] Mobley DL, Guthrie JP. FreeSolv: a database of experimental and calculated hydration free energies, with input files. J Comput Aided Mol Des. 2014;28:711-720. 10.1007/s10822-014-9747-x

[4] Duarte Ramos Matos G et al. Approaches for Calculating Solvation Free Energies and Enthalpies Demonstrated with an Update of the FreeSolv Database. J Chem Eng Data. 2017;62:1559-1569. 10.1021/acs.jced.7b00104

[5] Grossfield A et al. Best Practices for Quantification of Uncertainty and Sampling Quality in Molecular Simulations. Living J Comput Mol Sci. 2018;1:5067. 10.33011/livecoms.1.1.5067